

**ENGR 2303 Electrical Science**  
**Review Material for Test 1 and Sample Test**

Given Electric circuit diagram, you should be able to find the current, voltage, and power at any element in the circuit using any of the following techniques:

- a. Ohm's law with KVL and KCL
- b. Finding the equivalent of resistors connected in series and parallel.
- c. Nodal analysis,
- d. Loop (Mesh) analysis,
- e. Superposition.

**Question 1**

Find the power delivered to the 2- and 6-ohms resistors in Fig. 1.

**Question 2**

Find  $R_{eq}$  for the circuit shown in Fig. 2.

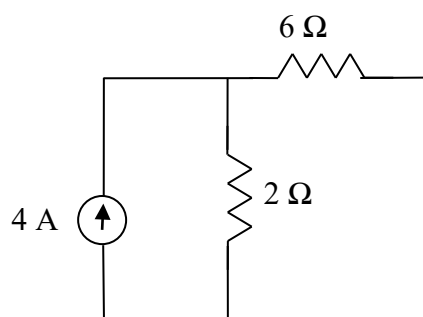


Fig. 1

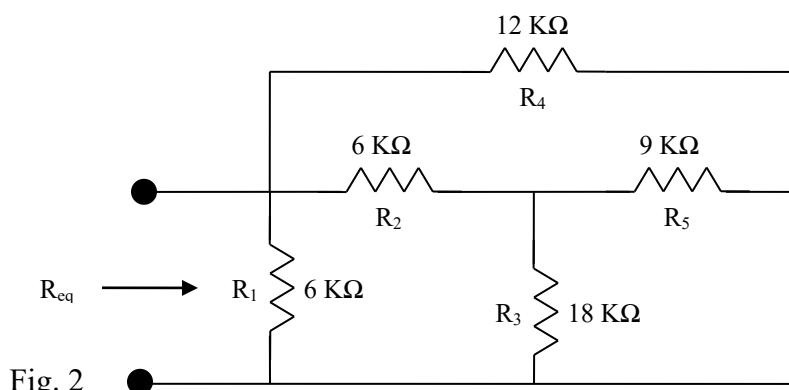


Fig. 2

**For questions 3 and 4 you need to solve them by two different methods (loop and nodal techniques).**

**Question 3**

Find  $I_o$  in the circuit in Fig. 3 below using super.

**Question 4**

Find  $V_o$  in the network in Fig. 4 below.

**Question 5:** Find  $I_o$  in the network in question 3 using the super position.

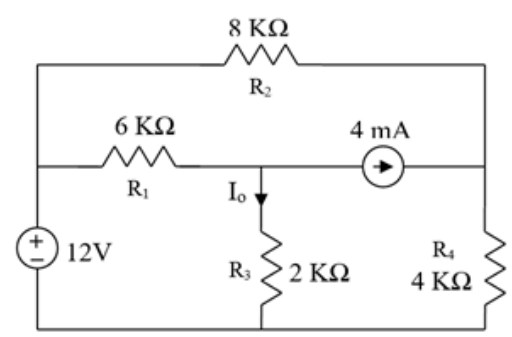


Fig. 3

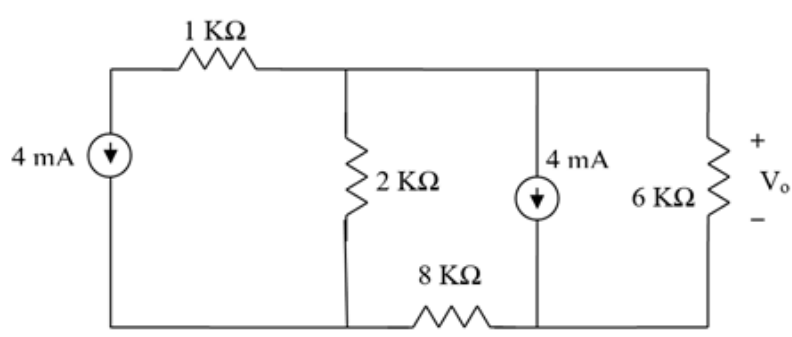


Fig. 4